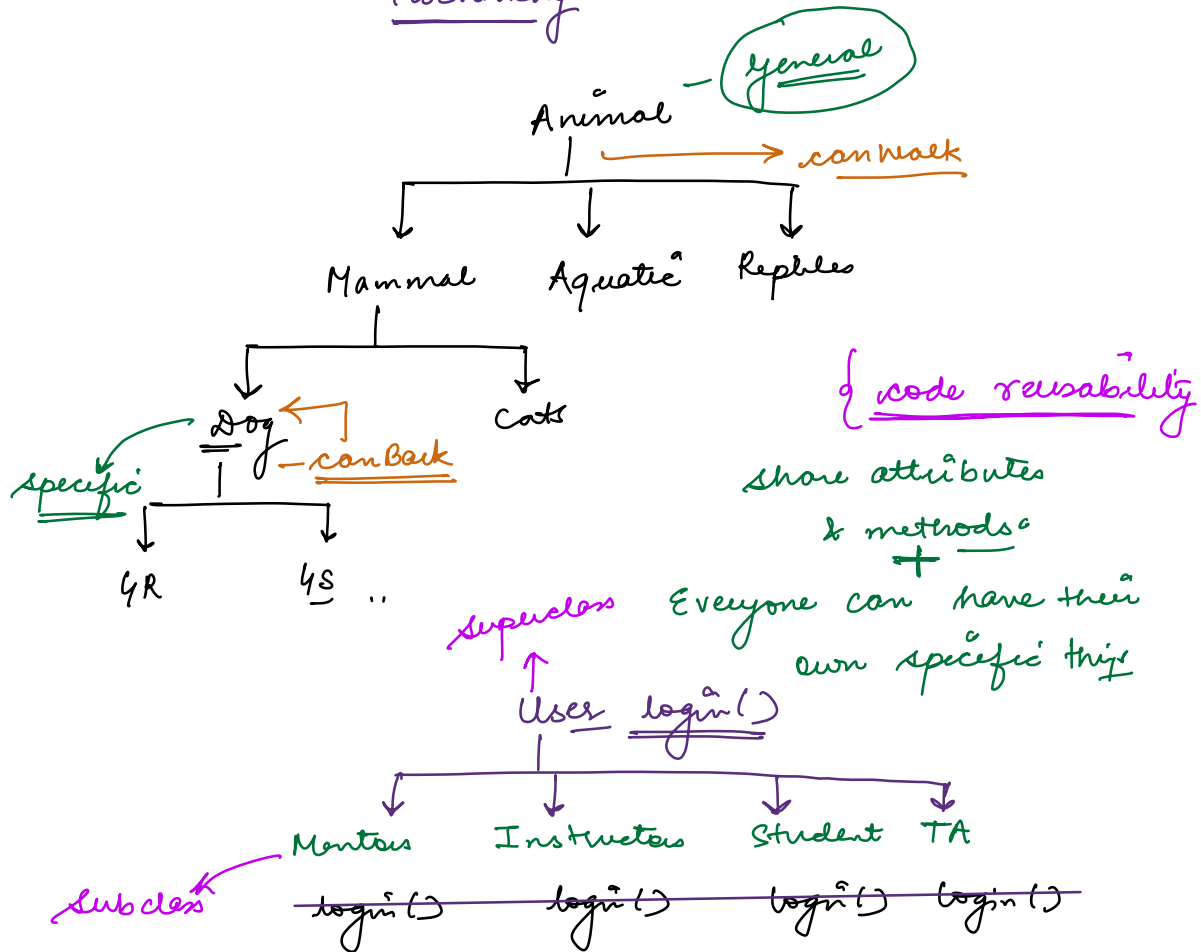


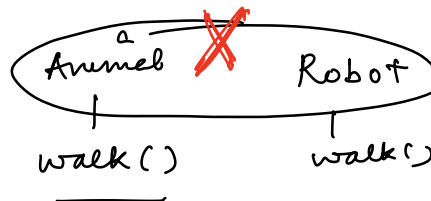
Inheritance & Polymorphism

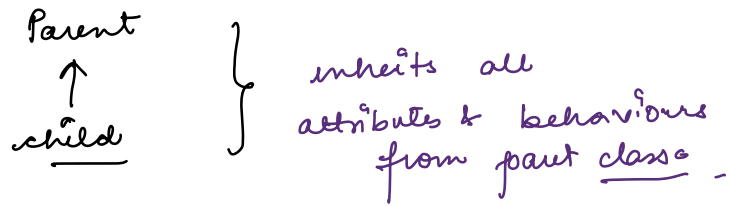
Inheritance

- Hierarchy



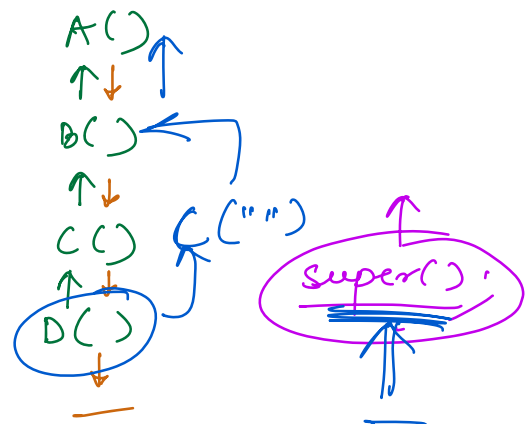
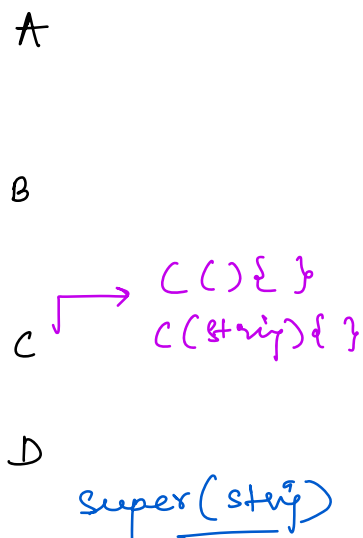
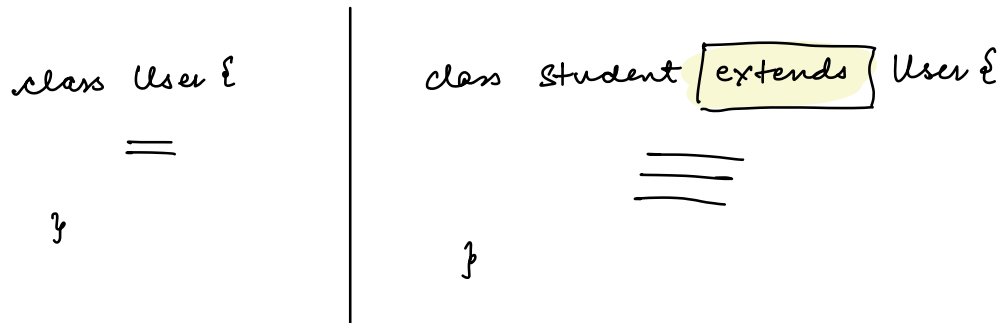
logically parent & child

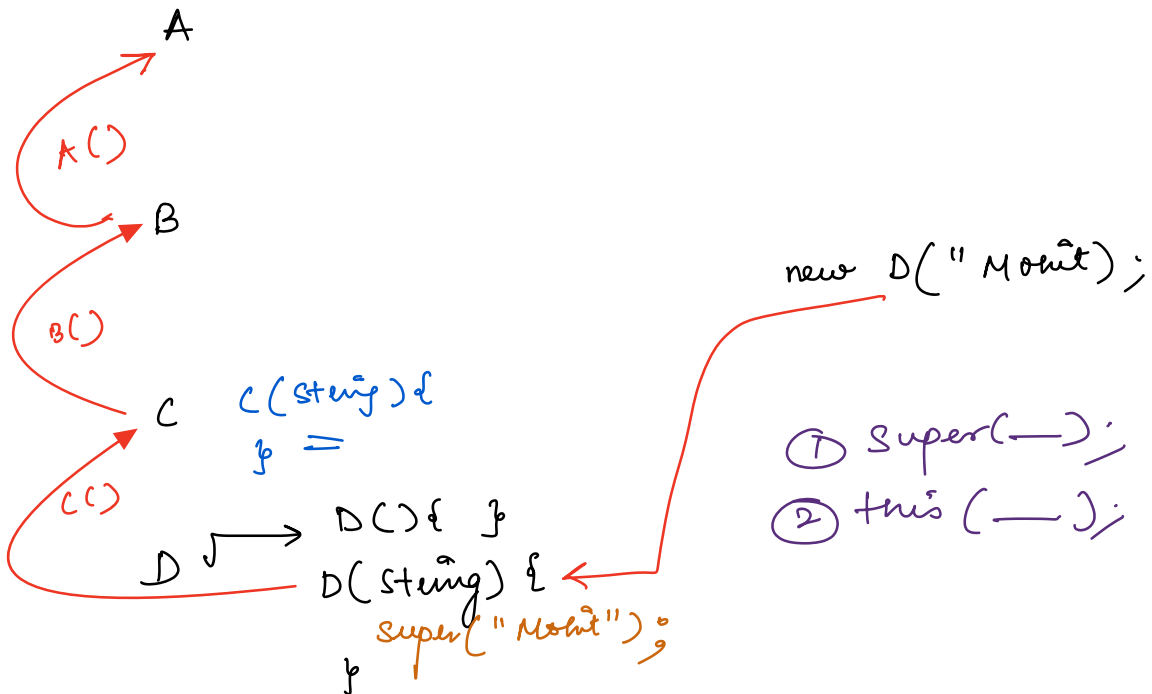
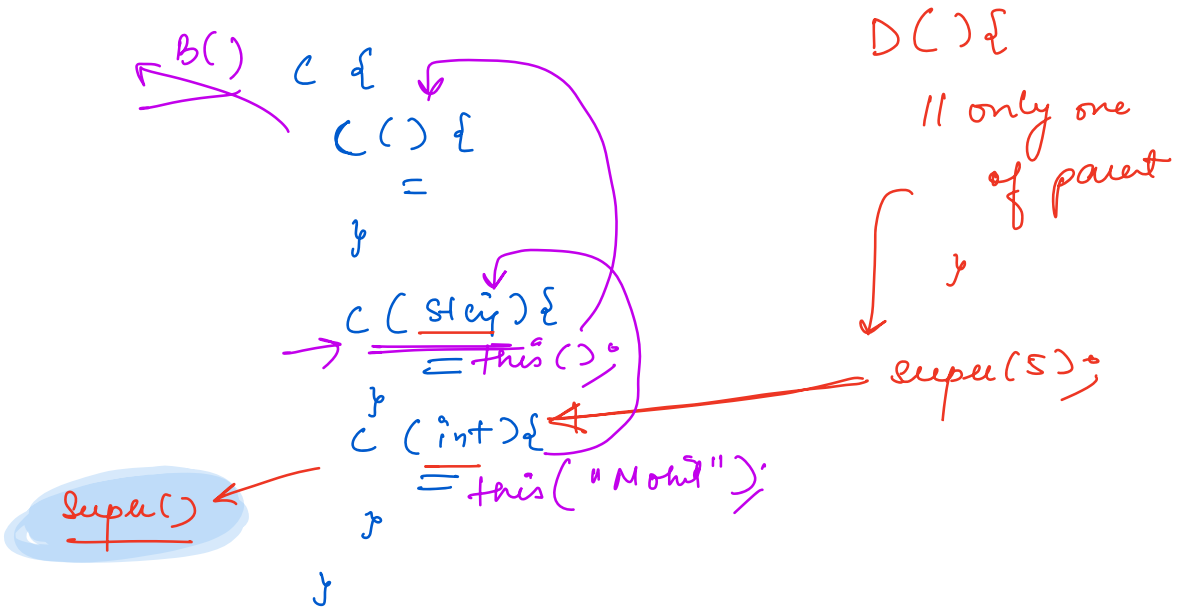




+ private members are not inherited

+ child may or may not add more members!

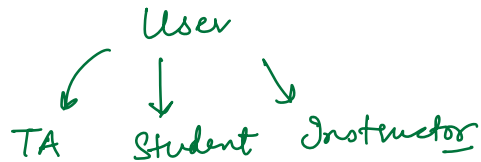




Poly morphism → forms.

many

someone who has many forms.



TA is a user
Student is a User
Instructor is a User

```
printUserNames( list < ? > users)
{
    _____ users.get(0) = name;
}
```

list < User > users;
↑
Student obj

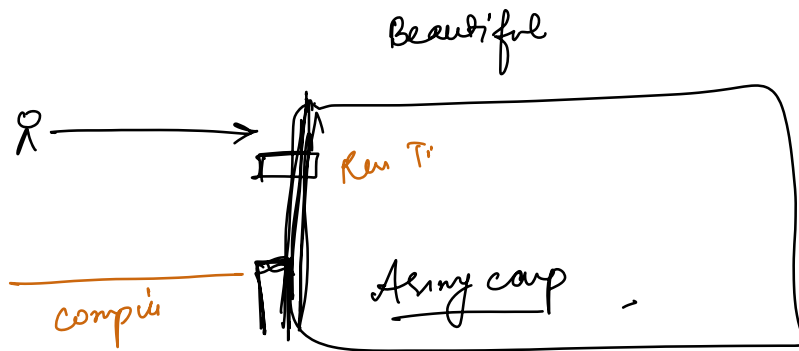
Parent class
ref variable
can point
towards child

User x = new Student();

vice versa is not allowed!

```
Dog d = new Dog();
d.bark();
```

✓ compile X Animal d = new Dog();
 X d.bark();
 d.walk();



utility.

```
changePassword(User x) {
    == user.password = " "
}
```

password.

User x = new User();

```
changePassword(new TAC());
strider();
```

Parent x = new child();

resolⁿ → x. _____ ;

↓

if parent has this ✓

class A {
 name;
 password;
}

extends A
class B {
 age;
}

extends B
class C {
 psp;
}

A a = ~~new C();~~ B();

a.name = _____;

a.age = _____;

compit error

a.psp = _____; ?

And a = new Dog();

✓ ~~Dog d = new Animal();~~

d.bark();

Method Overloading

class A {

① { hello() {
 }
}
② { hello(str) {
 ↑
 }
}

hello();
↑
hello("Mohit");

compile-Time

① void print()
 void print(str) ✓

② void print(str) ✓
 void print(str, int)

③ void print(int) ✓
 void print(double)

④ void print(int, str) ✓
 void print(str, int)

⑤ `void print (string)`
`void print (string)` 

⑥ `void print (string)`
`int` `print (string)` 
`print ("Hello")`

Method Signature

 `Name of Method (dataType1, dt2, dt3 ...)`

Method Overriding

class A {

① void^a doSomething(String) {
 ==
}

}

class B extends A {

String doSomething(String) { ~~XXXX~~
 ==
}

② void^a doSomething(String) {
 ==
}

}

Ⓐ a = new A();
 Ⓐ a.doSomething(); → ①

B b = new B();
b doSomething();

II

A a = new B();
a doSomething();

run-time
 polymorphism/
 overriding
=====

Parent x = new child();
 x = _____;

resolution
execution